

NONCOOPERATIVE GAME THEORY — ECE270: HOMEWORK #2

Exercise 3 (Alternate play). Consider the zero-sum matrix game defined by the matrix

$$A = \underbrace{\begin{bmatrix} -1 & -3 & -3 & 1 \\ 0 & 1 & -2 & -1 \\ 2 & -2 & 0 & -1 \end{bmatrix}}_{P_2 \text{ choices}} \Bigg\}_{P_1 \text{ choices},$$

in which the minimizer P_1 selects rows and the maximizer P_2 selects columns. Assume that P_1 plays first.

1. Find a pair of “rational” policies for each player.

Hint: The policy for P_2 should include “if” conditions.

2. Show that the pair of policies that you found does not lead to regret, in the sense that the policy of each player is the best response against the policy of the other. $\square$

Exercise 4 (Pure security levels/policies). The following matrices defines zero-sum matrix games for which both players have 4 and 3 actions, respectively,

$$A = \underbrace{\begin{bmatrix} 2 & -1 & 1 & -1 \\ -2 & -3 & 1 & -2 \\ -1 & -2 & -3 & -4 \\ 1 & -1 & 0 & -1 \end{bmatrix}}_{P_2 \text{ choices}} \Bigg\}_{P_1 \text{ choices}} \quad B = \underbrace{\begin{bmatrix} 1 & 1 & -1 \\ -2 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}}_{P_2 \text{ choices}} \Bigg\}_{P_1 \text{ choices}.$$

Compute the security levels, all security policies for both players, and all pure saddle-point equilibria (if they exist). $\square$

Exercise 5 (Multiple pure saddle-point). Give examples of 2×2 matrices that define zero-sum games with:

1. no pure saddle-point equilibria,
2. a single pure saddle-point equilibria,
3. just two distinct pure saddle-point equilibria,
4. just three distinct pure saddle-point equilibria,
5. just four distinct pure saddle-point equilibria.

Justify all of your answers.

Hint: some examples may not be possible. $\square$